API Documentation

Retail Transaction Analytics Platform

Backend: FastAPI | **Database:** MySQL | **Security:** JWT and role-based access control

1. Overview

The API serves the React/Vite dashboard using data in the MySQL *retail_analytics* database. It includes authentication, sales analytics, customer lookup, and product listing and management.

2. Start the Backend

Run from the project root (activate the project environment first, if used):

```
uvicorn src.api.main:app --reload
```

Local base URL: http://127.0.0.1:8000

3. API Endpoints

Group	Method	Endpoint	Purpose
Authentication	POST	/api/auth/login	Log in and obtain an access token.
Authentication	GET	/api/auth/me	Get the current authenticated user.
Analytics	GET	/api/analytics/summary	Return high-level retail summary metrics.
Analytics	GET	/api/analytics/monthly-sales	Return sales totals grouped by month.
Analytics	GET	/api/analytics/country-sales	Return sales grouped by country.
Analytics	GET	/api/analytics/top-products	Return top products; supports the limit query parameter.
Analytics	GET	/api/analytics/date-range	Return analytics for a date range; requires start_date and end_date.
Customers	GET	/api/customers	List customers with pagination.
Customers	GET	/api/customers/top	Return top customers; supports the limit query parameter.
Customers	GET	/api/customers/{customer_id}	Retrieve customer details by customer ID.
Products	GET	/api/products/	List products with optional search, sorting, and pagination.
Products	POST	/api/products/	Create a product; admin access required.
Products	GET	/api/products/{stock_code}	Retrieve a product by stock code.
Products	PUT	/api/products/{stock_code}	Update a product; admin access required.
Products	DELETE	/api/products/{stock_code}	Delete a product; admin access required.

4. Query Parameters

GET /api/analytics/top-products: *limit* controls how many top products are returned (example: *limit=10*).

GET /api/analytics/date-range: requires *start_date* and *end_date* to restrict analytics to a date interval.

GET /api/customers: *page* and *page_size* control pagination. **GET /api/customers/top**: *limit* controls the number of customers.

GET /api/products/: *search*, *sort_by*, *order*, *page*, and *page_size* support filtering, sorting, and pagination.

Example product request

```
GET /api/products/?search=bag&sort_by=revenue&order=desc&page=1&page_size=5
```

Returns the first five matching products sorted by revenue in descending order.

5. Authentication and Roles

For protected routes, send the token in the request header:

```
Authorization: Bearer <your_token>
```

Admin: can create, update, and delete products. **Analyst:** has read access to analytics and products.

6. Response Notes

Product listing returns pagination metadata and product records; an example shape used during development is:

```
{"page":1,"page_size":5,"total_products":5131,"products":[...]}
```

Analytics and customer endpoints return the metrics or records used by their corresponding dashboard views. Field-level response schemas are not reproduced here unless confirmed in the project notes.